Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/04/2024
PI Name:	Iyer, Janani Ph.D.		
Project Title:	Neurological, Cardiac, and Muscular Consequences of the Deep Space Environment in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	16
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBBLEO E.11: Research Pathfinder
Start Date:	11/01/2023	**End Date:**	10/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per NASA Ames Research Center (ARC), the following changes have been made to the list of Co-Investigators (Co-Is): Co-I change from April Ronca to Dr. Joshua Alwood, and 2) Dr. Karen Ocorr from collaborator to Co-I (Ed., 4/2/25).		
COI Name (Institution):	Vogler, Georg Ph.D. (Sanford Burnham Prebys Medical Discovery Institute) Mhatre, Siddhita Ph.D. (NASA Ames Research Center) Alwood, Joshua Ph.D. (NASA Ames Research Center) Ocorr, Karen (Sanford Burnham Prebys Medical Discovery Institute)		
Grant/Contract No.:	80NSSC24K0293		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	October 2024: Updated Task Description per the Principal Investigator (PI) Human space exploration and long-duration habitation in space necessitate a thorough understanding of the impact of the spaceflight environment on human physiology. This study aims to advance NASA's efforts by swiftly characterizing risks and identifying suitable countermeasures for future deep space missions. We hypothesize that exposure to spaceflight conditions, including altered gravity and radiation, induces oxidative stress responses, leading to behavioral and mitochondrial impairments, morphological changes in neural tissues, heart and muscle, and alterations in gene expression. Additionally, we propose that overexpression of an antioxidant gene, such as superoxide dismutase (SOD2), may mitigate these changes. This multi-system study will utilize Drosophila melanogaster to investigate the prolonged effects of spaceflight and post-flight acclimation. The fly model has exceptional genetic tools, requires limited laboratory maintenance, is well-established as a spaceflight model organism, and has been successfully flown by the Principal Investigator team to low-Earth orbit (LEO) using the proposed NASA Vented Fly Box (VFB) hardware. This investigation will use various assays to understand the molecular, morphological, functional, and behavioral alterations induced by space travel. Using different genetic fly lines, genetic, cellular, and functional changes will be characterized in space-reared flies and flies readapted to Earth conditions (R+10 days). This study will (a) focus on the cellular and molecular responses to oxidative stress and autophagy caused by microgravity and radiation in muscle, heart, and nervous system, (b) utilize fly lines with tissue-specific expression (neuronal, muscle, and cardiac) of an antioxidant gene, SOD (superoxide dismutase), and (c) use the well-characterized genetic mutant, Tor7/P, to test whether increased autophagy is beneficial or represents a maladaptive response to the spaceflight stressors. Data from this mission will also be compared with previous LEO-based data to identify shared signatures. Furthermore, cross-species analysis of the transcriptomic data will be performed to compare this mission to other invertebrate and vertebrate spaceflight-related data to determine the evolutionarily conserved pathways perturbed by space stressors across species. This project aims to provide crucial insights into the mechanisms underlying oxidative stress responses, synaptic changes, and heart and muscle deficits, facilitating the identification of diagnostic and therapeutic targets to mitigate the adverse health effects of long-duration space habitation. Ultimately, this research will contribute to our ability to thrive in deep space and inform future missions to the Moon and Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	October 2024 Update per PI Potential Impact: The experiments proposed here will test the hypothesis that oxidative stress induced by spaceflight stressors negatively impacts the CNS, heart, and somatic muscles and that tissue-specific overexpression of an antioxidant gene such as superoxide dismutase, can mitigate these effects. This study will use different genetic lines to characterize the genetic, cellular, and functional changes in space-reared flies and their adaptation to Earth's environment across multiple organ systems. We predict sex-specific differences in neuronal, cardiac, and muscular outcomes, informing distinct male-female responses to spaceflight exposure. Detailed behavioral, structural, and multi-omics data from brain, muscle, and heart tissue obtained from the same animals -- permitting an integrated assessment of the effects of the space environment across tissues.
Task Progress:	To explore the effects of spaceflight and Earth readaptation on the neural, cardiac, and somatic muscle of Drosophila (Aims 1 and 2), we plan to send several Drosophila strains to the International Space Station (ISS): fly lines with tissue-specific expression (neural, muscle, and cardiac) of the antioxidant gene superoxide dismutase 2, (SOD2), and a well-characterized genetic mutant, Tor7/P, to test whether increased expression of an antioxidant gene (SOD2) or autophagy is beneficial or represent maladaptive responses to spaceflight stressors. In the past year, we: 1) Revised the specific aims to account for the change in the mission from Artemis II to ISS 2) Reviewed and submitted the Science Requirements Document (SRD) 3) Reviewed and finalized the Advanced Transport Operating Systems (ATOPs) for flight 4) Finalized the food recipe for the flight 5) Finished Pre-Science Verification Test (SVT) experiments - Finalized volume of food - Finalized seeding number of flies - Generated fly lines with Sod2 overexpression in central nervous system (CNS), muscles, and heart 6) Completed Science Verification Test Readiness Review (SVTRR) 7) Calibrated incubator for SVT and Experiment Verification Test (EVT) 8) Planning and ordering of EVT supplies (chemicals and reagents for fly food, vials, bottles, vial and bottle flugs, Fly pads, and flow buddy) 9) Conducted SVT at NASA Ames Research Center (Ames) and Sanford Burnham Prebys Medical Discovery Institute (SBP) o Egg lay scenario with nominal (31 days) and worst case (38 days) timeline o Adults scenario with nominal and worst case timeline o Using the NASA Vented Fly Box (VFB) in different orientations to accommodate different types of fly vials along with controls inside the incubator 10) Completed SVT at Ames and SBP o Compared and analyzed the SVT data from both institutions o A few transgenic lines were not as healthy and did not yield as many flies, so we planned to conduct a delta SVT to finalize the conditions before the start of EVT 11) Ongoing delta SVT with 4 different scenarios o Egg lay with 48 hr incubation at 30C followed by 22C for the timeline of the mission o Egg lay followed by 22C for the timeline of the mission o Adult flies with 48 hr incubation in 30C followed by 22C for the timeline of the mission o Adult flies followed by 22C for the timeline of the mission
Bibliography Type:	Description: (Last Updated: 04/08/2025)
Abstracts for Journals and Proceedings	Iyer J, Tasoula A, Costes SV, Ronca AE, Szewczyk NJ, Vogler G, Ocorr K, Mhatre SD. "iGCE and MitoFlyght spaceflight missions: Unraveling oxidative stress responses in space. " 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024